THzSim2 Measured-Reference Drude Study Notes

THzSim2 Repository

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1 Purpose

This note documents the equations used by the measured-reference compatibility workflow in `thzsim2`. The intended use is to explain the code path behind the one-layer Drude study that starts from a measured reference CSV, resamples it to a uniform time grid, fits a single Drude layer, and exports both the old-compatible mean-squared-error summary and the new diagnostic metrics.

2 Reference Trace Import and Resampling

The canonical time-trace CSV schema is

`time_ps,trace`

but the importer also accepts common measured-data headers such as

`Time_abs/ps, Signal/nA`.

If the measured time axis is not exactly uniform, the workflow preserves the raw amplitude values but interpolates them onto a uniform grid. Let the imported samples be

$$\{(t_i, E_i)\}_{i=0}^{N-1}, \qquad t_{i+1} > t_i.$$

The dominant or median time step is used as the target spacing

$$\Delta t_{\text{resamp}} = \text{median}(t_{i+1} - t_i),$$

and the uniform grid is

$$\tilde{t}_i = t_0 + i\,\Delta t_{\text{resamp}}, \qquad i = 0, \ldots, N-1.$$

The resampled trace is then

$$\tilde{E}(\tilde{t}_i) = \text{interp}\big(\tilde{t}_i; \{t_j, E_j\}\big)\,,$$

using linear interpolation.

The compatibility workflow keeps the imported amplitude in source units. In the example measured CSV this means the exported reference trace remains in nA rather than being normalized.

3 Fourier Transform Convention

The reference and sample traces are propagated through the frequency domain using the discrete Fourier-transform convention implemented in `thzsim2.core.fft`. If the time trace is

$$E_n = E(t_n), \qquad t_n = t_0 + n\Delta t,$$

then the angular-frequency grid is

$$\omega_k = 2\pi f_k,$$

with the standard FFT ordering from the uniform grid. The forward transform in code is

$$\hat{E}(\omega_k) = \Delta t\, N\ \mathrm{IFFT}(E_n)\, e^{i\omega_k t_0},$$

and the inverse transform is

$$E_n = \frac{1}{N\Delta t}\,\mathrm{FFT}\Big(\hat{E}(\omega_k)e^{-i\omega_k t_0}\Big).$$

This convention preserves the original time-origin offset t_0, which matters when the reference trace is imported from a measured file rather than generated symmetrically around zero.

4 Single-Layer Transmission Model

For normal incidence, the complex field transmission through one layer of thickness d, refractive index $n_1(\omega)$, input medium n_0, and output medium n_2 is

$$T(\omega) = t_{01}(\omega)\, p_1(\omega)\, t_{12}(\omega)\, F_{\mathrm{FP}}(\omega),$$

where the Fresnel coefficients are

$$t_{jk} = \frac{2n_j}{n_j + n_k}, \qquad r_{jk} = \frac{n_j - n_k}{n_j + n_k},$$

the propagation factor is

$$p_1(\omega) = \exp\left(i\omega n_1(\omega)\frac{d}{c_0}\right),$$

and the finite Fabry–Perot correction used by the code is

$$F_{\mathrm{FP}}(\omega) = \sum_{m=0}^{M} \left[r_{12}(\omega)p_1(\omega)r_{10}(\omega)p_1(\omega)\right]^m.$$

Here M is the configured maximum number of internal reflections. In the old-compatible study the default is $M = 8$.

5 Drude Material Model

The single fitted layer uses the Drude dielectric function

$$\varepsilon(\omega) = \varepsilon_\infty - \frac{\omega_p^2}{\omega(\omega + i\gamma)}.$$

The complex refractive index is obtained from

$$\tilde{n}(\omega) = n(\omega) + ik(\omega) = \sqrt{\varepsilon(\omega)},$$

with the passive branch selected so that

$$k(\omega) \geq 0.$$

The public notebook API uses THz for both the Drude plasma frequency and damping:

$$\omega_p = 2\pi f_p \times 10^{12}, \qquad \gamma = 2\pi f_\gamma \times 10^{12}.$$

5.1 Conversion from Conductivity and Relaxation Time

The compatibility study follows the old script and often describes true values in terms of the DC conductivity σ and relaxation time τ. The code converts these to Drude parameters by

$$\gamma = \frac{1}{\tau},$$

$$\sigma = \varepsilon_0 \frac{\omega_p^2}{\gamma},$$

so that

$$\omega_p = \sqrt{\frac{\sigma}{\varepsilon_0 \tau}}.$$

The notebook-facing helper functions therefore implement

$$f_\gamma = \frac{1}{2\pi\tau}, \qquad f_p = \frac{1}{2\pi \times 10^{12}} \sqrt{\frac{\sigma}{\varepsilon_0 \tau}}.$$

6 Why the Exported n, k CSV Starts Above Zero Frequency

For a Drude medium,

$$\varepsilon(\omega) = \varepsilon_\infty - \frac{\omega_p^2}{\omega(\omega + i\gamma)}.$$

As $\omega \to 0$, the conductivity-driven response becomes singular, so the apparent $n(\omega)$ and $k(\omega)$ can become very large. The code still handles the $\omega \approx 0$ behavior internally through a protected frequency floor, but the exported sample-layer CSVs and plots omit the exact 0 THz row so that the saved `freq_thz,n,k` files remain numerically readable and visually useful.

7 Noise Model

The compatibility study adds white Gaussian noise to the clean simulated sample trace. If the requested dynamic range is D dB and the peak absolute sample amplitude is

$$A_{\max} = \max_t |E_{\text{sample,clean}}(t)|,$$

then the noise standard deviation is

$$\sigma_{\text{noise}} = \frac{A_{\max}}{10^{D/20}}.$$

The observed synthetic sample trace is

$$E_{\text{obs}}(t) = E_{\text{sample,clean}}(t) + \eta(t), \qquad \eta(t) \sim \mathcal{N}(0, \sigma_{\text{noise}}^2).$$

8 Objective Functions and Diagnostic Metrics

8.1 Raw Mean-Squared Error

For compatibility with the old study, the primary fit objective remains the raw mean-squared error

$$\text{MSE} = \frac{1}{N} \sum_{i=0}^{N-1} \left(E_{\text{model}}(t_i) - E_{\text{obs}}(t_i)\right)^2.$$

This value can be numerically small when the observed trace amplitude itself is small, so it must be interpreted together with the trace plots and the scale-aware diagnostics below.

8.2 Relative L^2 Error

The relative L^2 metric exported by the workflow is

$$\text{relative_}L^2 = \frac{\|E_{\text{model}} - E_{\text{obs}}\|_2}{\max(\|E_{\text{obs}}\|_2, 10^{-30})}.$$

8.3 Windowed MSE

The added diagnostic `windowed_mse` focuses on the part of the waveform where the observed signal is not negligible. With a threshold fraction α (currently $\alpha = 0.05$),

$$\mathcal{I} = \left\{ i : |E_{\text{obs}}(t_i)| \geq \alpha \max_j |E_{\text{obs}}(t_j)| \right\}.$$

Then

$$\text{windowed_MSE} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \left(E_{\text{model}}(t_i) - E_{\text{obs}}(t_i) \right)^2.$$

If no samples pass the threshold, the code falls back to the raw MSE.

8.4 Peak-Normalized RMSE

To make the residual scale easier to interpret visually, the workflow also exports

$$\text{peak_normalized_RMSE} = \frac{\sqrt{\frac{1}{N} \sum_i \left(E_{\text{model}}(t_i) - E_{\text{obs}}(t_i) \right)^2}}{\max(\max_i |E_{\text{obs}}(t_i)|, 10^{-30})}.$$

9 Optimization Strategy

The fitting code uses a two-stage optimization strategy:

1. a global differential-evolution search over the bounded fit variables;
2. a local `L-BFGS-B` refinement starting from the global best point.

For the old-compatible study, the bounded fit variables are:

Parameter	Initial value	Lower bound	Upper bound
Thickness	150 µm	40 µm	320 µm
τ	3 ps	0.05 ps	30 ps
σ	$0.01\,\mathrm{S\,m^{-1}}$	$5 \times 10^{-4}\,\mathrm{S\,m^{-1}}$	$0.2\,\mathrm{S\,m^{-1}}$

Internally, the optimizer works on thickness, Drude plasma frequency, and Drude damping frequency, because those are the native variables of the forward model.

10 Parameter-Uncertainty Estimate

After the optimum $\hat{\mathbf{x}}$ is found, the code estimates local parameter uncertainties with a finite-difference Jacobian of the residual vector

$$\mathbf{r}(\mathbf{x}) = E_{\text{obs}} - E_{\text{model}}(\mathbf{x}).$$

The Jacobian is approximated column-by-column as

$$J_{:,j} \approx \frac{\mathbf{r}(\hat{\mathbf{x}} + h_j \mathbf{e}_j) - \mathbf{r}(\hat{\mathbf{x}} - h_j \mathbf{e}_j)}{2h_j},$$

with a bounded step size chosen from the parameter magnitude and the parameter span.

If there are m residual samples and p fitted parameters, the residual variance estimate is

$$\hat{\sigma}^2 = \frac{\mathbf{r}^\mathsf{T}\mathbf{r}}{\max(m-p,1)}.$$

The covariance estimate is then

$$\Sigma \approx \hat{\sigma}^2 (J^\mathsf{T} J)^+,$$

where $(\cdot)^+$ denotes the pseudoinverse. The reported parameter standard deviations are

$$\sigma_j = \sqrt{\Sigma_{jj}},$$

and the reported parameter correlations are

$$\rho_{ij} = \frac{\Sigma_{ij}}{\sigma_i \sigma_j}.$$

11 Progress, ETA, and Study Size

The old-compatible workflow evaluates a pilot set of cases spread across the full sweep, estimates the average runtime per case, and prints an initial total runtime estimate before the full study loop begins. For the requested sweep

$$4 \times 20 \times 20 \times 3 = 4800$$

cases are run, with one replicate per case.

12 Known Pitfalls

- **Pulse placement matters.** A generated pulse can look artificially tiny if its center lies outside the generated time window. The notebook-facing pulse generator now rejects that configuration explicitly.
- **Measured traces can carry baseline and DC offset.** The compatibility workflow keeps the measured amplitude scale and baseline for apples-to-apples comparison, but it records basic baseline diagnostics in the reference manifest.
- **Raw MSE is scale-dependent.** A visibly poor fit can still have a small raw MSE if the trace amplitude is small. The workflow therefore exports `windowed_mse`, `relative_l2`, and `peak_normalized_rmse` alongside the raw MSE.
- **Drude DC export is not plot-friendly.** The internal model still handles the low-frequency limit, but the exported `freq_thz,n,k` files start at the first positive frequency to avoid an unreadable DC row.